

Homework 3

4.11. Study the state of equilibrium between a gaseous phase and an adsorbed phase in a single-component system. Show that the pressure in the gaseous phase is given by the Langmuir equation

$$P_g = \frac{\theta}{1-\theta} \times (\text{a certain function of temperature}),$$

where θ is the equilibrium fraction of the adsorption sites that are occupied by the adsorbed molecules.

Goal : Find P_g for the system in the problem.

Solution

Using the solution from 4.10, we find that the chemical potential μ of the system is

$$\mu = kT \ln \frac{N}{(N_0 - N)a(T)}$$

where N : no. of molecules adsorbed

N_0 : no. of adsorption sites

$a(T)$: partition function of single molecule

Since θ is the fraction of adsorption sites occupied $\therefore \theta = \frac{N}{N_0}$,

$$\mu = kT \ln \frac{N/N_0}{(1 - N/N_0)a(T)} = kT \ln \frac{\theta}{(1-\theta)a(T)}$$

Using our results from class for an ideal gas :

$$\bullet \quad Z = e^{\mu/kT} \quad (4.3.11) \quad (1)$$

$$\bullet \quad \phi(T) \approx 1/Z \quad (4.4.26) \quad (2)$$

$$\bullet \quad P_v = kT \frac{f(T)}{\phi(T)} \quad (4.4.28) \quad (3)$$

$$\bullet \quad f(T) = (2\pi m kT)^{3/2} / h^3 \quad (4.4.29) \quad (4)$$

We have

$$Z = e^{\mu/kT} = \frac{\theta}{(1-\theta)a(T)} \quad (1)$$

$$\phi(T) \approx \frac{(1-\theta)a(T)}{\theta} \quad (2)$$

$$P_v \stackrel{(3)}{=} kT \frac{f(T)}{\phi(T)} \stackrel{(4)}{=} kT \frac{(2\pi m kT)^{3/2} / h^3}{\frac{1-\theta}{\theta} a(T)} = \frac{\theta}{1-\theta} \times h(T) \quad \square$$

- 4.14. Assuming that the latent heat of vaporization of water $L_v = 2260 \text{ kJ/kg}$ is independent of temperature and the specific volume of the liquid phase is negligible compared with the specific volume of the vapor phase, $v_{\text{vapor}} = kT/P_o(T)$, integrate the Clausius–Clapeyron equation (4.7.7) to obtain the coexistence pressure as a function of temperature. Compare your result with the experimental vapor pressure of water from the triple point to 200°C . The equilibrium vapor pressure at 373 K is $101 \text{ kPa} = 1 \text{ atm}$.

Goal Find $P_o(T)$ by solving the Clausius–Clapeyron Equation.

Solution

Assumptions

- Latent Heat of vaporization of water: $L_v = 2260 \text{ kJ/kg}$ (doesn't depend on T)
- $v_{\text{liquid}} \ll v_{\text{vapor}}$
- specific volume of vapor: $v_{\text{vapor}} = \frac{kT}{P_o(T)}$

Clausius–Clapeyron Equation: (4.6.7)

$$\frac{dP_o}{dT} = \frac{L}{T \Delta v} \quad \text{or} \quad \frac{L_v}{T(v_{\text{vapor}} - v_{\text{liquid}})}$$

Using assumptions $v_{\text{vapor}} - v_{\text{liquid}} \hat{=} v_{\text{vapor}}$ & $v_{\text{vapor}} = \frac{kT}{P_o(T)}$

$$\frac{dP_o}{dT} = \frac{L_v}{T \cdot \frac{kT}{P_o(T)}} = \frac{L_v}{kT^2} P_o(T)$$

Separation of variables yields:

$$\frac{1}{P_o(T)} dP_o = \frac{L_v}{kT^2} dT$$

$$\ln(P_o(T)) = -\frac{L_v}{kT} + C$$

$$P_o(T) = C \exp\left[-\frac{L_v}{kT}\right]$$

Invoking initial condition $P_o(T_0) = P_0$

$$P_0 = C e^{-L_v/kT_0} \Rightarrow C = P_0 e^{L_v/kT_0}$$

Substituting in C

$$P_o(T) = P_0 \exp\left[-\frac{L_v}{k}\left(\frac{1}{T} - \frac{1}{T_0}\right)\right]$$

Let's confirm our results with experimentation taking water as an example.

Equilibrium vapor pressure of water: $P_0 = 101 \text{ kPa}$, $T_0 = 373 \text{ K}$

Latent Heat L_v per molecule in SI units: $L_v = 2260 \frac{\text{kJ}}{\text{kg}} \cdot \underbrace{10^3 \frac{\text{J}}{\text{kJ}}}_{\text{kJ} \rightarrow \text{J conversion}} \cdot \underbrace{10^{-3} \frac{\text{kg}}{\text{g}}}_{\text{kg} \rightarrow \text{g conv.}} \cdot \underbrace{18.015 \frac{\text{g}}{\text{mol}}}_{\text{molecular mass}} \cdot \underbrace{\frac{1}{6.023 \times 10^{23}} \frac{\text{mol}}{\text{molecule}}}_{\text{Avogadro's constant}}$

$$\approx 6.760 \times 10^{-20} \text{ J/molecule}$$

Boltzmann's constant: $k \approx 1.381 \times 10^{-23} \frac{\text{m}^2 \text{kg}}{\text{s}^2 \text{K}}$

Putting all of our constants in $P_0(T)$

$$P_0(T) = 101 \text{ kPa} \cdot \exp \left\{ - \frac{6.760 \times 10^{-20}}{1.381 \times 10^{-23}} \left(\frac{1}{T} - \frac{1}{373} \right) \right\}$$

$$\approx 101 \exp \left\{ -4894 \left(\frac{1}{T} - \frac{1}{373} \right) \right\} \text{ kPa}$$

From the Engineering Toolbox website, we can list the experimental P_0 at various different temperatures.

Temperature ($^{\circ}\text{C}$)	Experimental P_0 (kPa)	Theoretical P_0 (kPa)
0.01°C (triple point)	0.61165	0.8349
25°C (Room temp)	3.1699	3.7479
100°C (Boiling temp)	101.42	101.53
200°C (Asked max temp)	1554.9	1623.47

We find that our approximation becomes worse at higher temperatures. That may be due to our assumptions getting violated for large T .

- L_v may depend on T .

- $v_{\text{vapor}} \neq \frac{kT}{P_0}$

- $v_{\text{liquid}} \neq v_{\text{vapor}}$

- 4.15. Assuming that the latent heat of sublimation of ice $L_s = 2834 \text{ kJ/kg}$ is independent of temperature and the specific volume of the solid phase is negligible compared with the specific volume of the vapor phase, $v_{\text{vapor}} = kT/P_\sigma(T)$, integrate the Clausius–Clapeyron equation (4.7.7) to obtain the coexistence pressure as a function of temperature. Compare your result to the experimental vapor pressure of ice from $T = 0$ to the triple point. The equilibrium vapor pressure at the triple point is 612 Pa.

Goal : Compare the coexistential pressure of ice P_σ between theory & experiment.

Equilibrium vapor pressure of ice : $P_\sigma = 612 \text{ kPa}$, $T_\sigma = 273.15 \text{ K}$

Latent Heat L_v per molecule in SI units : $L_v = 2834 \frac{\text{kJ}}{\text{kg}} \cdot \underbrace{10^3 \frac{\text{J}}{\text{kJ}}}_{\text{kJ} \rightarrow \text{J conversion}} \cdot \underbrace{10^{-3} \frac{\text{kg}}{\text{g}}}_{\text{kg} \rightarrow \text{g conv.}} \cdot \underbrace{18.015 \frac{\text{g}}{\text{mol}}}_{\text{molecular mass}} \cdot \underbrace{\frac{1}{6.023 \times 10^{23}} \frac{\text{mol}}{\text{molecule}}}_{\text{Avogadro's constant}}$

$$\approx 8.477 \times 10^{-20} \text{ J/molecule}$$

Boltzmann's constant : $k = 1.381 \times 10^{-23} \frac{\text{m}^2 \text{kg}}{\text{s}^2 \text{K}}$

Putting all of our constants in $P_\sigma(T)$

$$P_\sigma(T) = 612 \text{ kPa} \cdot \exp \left\{ - \frac{8.477 \times 10^{-20}}{1.381 \times 10^{-23}} \left(\frac{1}{T} - \frac{1}{273.15} \right) \right\}$$

$$\approx 612 \exp \left\{ - 6138 \left(\frac{1}{T} - \frac{1}{273.15} \right) \right\} \text{ kPa}$$

Using a paper titled Vapor Pressure Formulation for Ice p.17, we list the experimental P_σ for ice.

Temperature ($^\circ\text{C}$)	Experimental P_σ (kPa)	Theoretical P_σ (kPa)
0°C	611.153	612
-10°C	259.922	260.55
-20°C	103.276	103.69
-50°C	3.94017	3.9818

We conclude that the results match experimentation as our assumption hold at these temps.

- 4.16. Calculate the slope of the solid-liquid transition line for water near the triple point $T = 273.16 \text{ K}$, given that the latent heat of melting is 80 cal/g , the density of the liquid phase is 1.00 g/cm^3 , and the density of the ice phase is 0.92 g/cm^3 . Estimate the melting temperature at $P = 100 \text{ atm}$.

Goal : Calculate $\frac{dP}{dT}$ for water near the triple point $T = 273.16 \text{ K}$

Solution

Latent heat of melting : $L_s = 80 \frac{\text{cal}}{\text{g}} \cdot \frac{4.184 \text{ J}}{1 \text{ cal}} = 334.72 \text{ J/g}$

Specific volume at liquid phase : $v_{\text{liquid}} = \frac{1}{\rho_{\text{liquid}}} = 1 \text{ cm}^3/\text{g}$

Specific volume at ice phase : $v_{\text{ice}} = \frac{1}{\rho_{\text{ice}}} = 1.087 \text{ cm}^3/\text{g}$

Clausius - Clapeyron Equation : (4.6.7)

$$\frac{dP}{dT} = \frac{L}{T \Delta v} \quad \text{or} \quad \frac{L_s}{T(v_{\text{liquid}} - v_{\text{ice}})}$$

$$= \frac{334.72 \text{ J/g}}{273.16 \text{ K} \cdot (1 - 1.087) \text{ cm}^3/\text{g} \cdot 10^{-6} \text{ m}^3/\text{cm}^3}$$

$$\approx -1.41 \times 10^7 \text{ Pa/K}$$

Now to estimate the melting point at $P = 100 \text{ atm}$, we can compare it to the melting point at $P = 1 \text{ atm}$ & $T = 273.16 \text{ K}$.

Using a linear approximation :

$$\Delta P \approx \frac{dP}{dT} \Delta T$$

$$P_{100 \text{ atm}} - P_{1 \text{ atm}} \approx \frac{dP}{dT} (T_{100 \text{ atm}} - T_{1 \text{ atm}})$$

$$(100 - 1) \text{ atm} \approx 1.41 \times 10^7 \text{ Pa/K} \cdot \frac{1 \text{ atm}}{101325 \text{ Pa}} \cdot (T_{100 \text{ atm}} - 273.16)$$

$$T_{100 \text{ atm}} \approx -0.714 + 273.16$$

$$\approx 272.45 \text{ K} \approx -0.714^\circ \text{C}$$

This result implies that as pressure is added, the melting point gets lower i.e.

pressure $\xrightarrow{\text{induces}}$ melting.

4.17. Show that the Clausius–Clapeyron equation (4.7.7) guarantees that each of the coexistence curves at the triple point of a material “points into” the third phase; for example, the slope of the solid–vapor coexistence line has a value in-between the slopes of the solid–liquid and liquid–vapor coexistence lines.

Example:



Goal Show that $\left(\frac{dP}{dT}\right)_{SV}$ is between $\left(\frac{dP}{dT}\right)_{SL}$ and $\left(\frac{dP}{dT}\right)_{LV}$

Solution

Using the Clausius – Clapeyron Equation (4.6.7), we can express

$$\left(\frac{dP}{dT}\right)_{SV} = \frac{\Delta S_{SV}}{\Delta V_{SV}}$$

$$\left(\frac{dP}{dT}\right)_{SL} = \frac{\Delta S_{SL}}{\Delta V_{SL}}$$

$$\left(\frac{dP}{dT}\right)_{LV} = \frac{\Delta S_{LV}}{\Delta V_{LV}}$$

Invoking the constraint at the triple point (4.6.8)

$$\begin{aligned} \Delta S_{SV} + \Delta S_{SL} + \Delta S_{LV} &= 0 \quad \Rightarrow \quad \Delta S_{SV} = -\Delta S_{SL} - \Delta S_{LV} \\ \Delta V_{SV} + \Delta V_{SL} + \Delta V_{LV} &= 0 \quad \Rightarrow \quad \Delta V_{SV} = -\Delta V_{SL} - \Delta V_{LV} \end{aligned}$$

Rewriting $\left(\frac{dP}{dT}\right)_{SV}$

$$\begin{aligned} \left(\frac{dP}{dT}\right)_{SV} &= \frac{\Delta S_{SV}}{\Delta V_{SV}} = \frac{-\Delta S_{SL} - \Delta S_{LV}}{-\Delta V_{SL} - \Delta V_{LV}} \\ &= \frac{\Delta V_{SL} \left(\frac{dP}{dT}\right)_{SL} + \Delta V_{LV} \left(\frac{dP}{dT}\right)_{LV}}{\Delta V_{SL} + \Delta V_{LV}} \end{aligned}$$

If we were to take $\frac{\Delta V_{SL}}{\Delta V_{SL} + \Delta V_{LV}} = \alpha$. Then, we can rewrite

$$\left(\frac{dP}{dT}\right)_{SV} = \alpha \left(\frac{dP}{dT}\right)_{SL} + (1-\alpha) \left(\frac{dP}{dT}\right)_{LV}$$

Since typically $\Delta V_{SL}, \Delta V_{LV} > 0$, then by Jensen's Inequality $\left(\frac{dP}{dT}\right)_{SV}$ is between both $\left(\frac{dP}{dT}\right)_{SL}$ & $\left(\frac{dP}{dT}\right)_{LV}$